



Signal processing and calibration of a low-cost inductive rain sensor for raindrop detection and precipitation calculation

Christoph Clemens^{a,*}, Annette Elisabeth Jobst^a, Mario Radschun^a, Jörg Himmel^a,
Olfa Kanoun^b

^a Institute for Measurement Engineering and Sensor Technology, Ruhr West University of Applied Sciences, Duisburger Str. 100, 45479 Mülheim an der Ruhr, Germany

^b Chair for Measurement and Sensor Technology, Chemnitz University of Technology, Reichenhainer Str. 70, 09126 Chemnitz, Germany

ARTICLE INFO

Keywords:

Smart sensor
Rain sensor
Signal processing
Damped harmonic detection
Drop size determination
Precipitation calculation

ABSTRACT

Precipitation measurements in urban areas are becoming more and more important due to the increasing number of heavy rainfall events. In order to establish a dense measurement network within a city, low-cost sensors are necessary. A low-cost inductive rain sensor was developed for this purpose. In this study, the signal processing of the sensor is presented. To collect measurement data, the sensor was deployed in the field next to a high-precision reference sensor. An algorithm has been designed to detect the drops falling on the sensor. Subsequently, two methods have been developed to determine the amount of precipitation from the detected drops. The first approach is based on the statistical drop size distribution, which can be described by a gamma function. In the second approach, drop sizes are assigned to the detected drops based on their peak height. The drop detection algorithm has an accuracy of 89.25 %. The two methods for precipitation calculation have deviations of 11.11 % and 11.51 % from the reference measurement.

1. Introduction

Precise and locally resolved precipitation measurement data is of key importance in many areas of application. These include weather forecasting, climate change, climate research, agriculture and water management. Precipitation measurements can be used to develop strategies for preparedness and recommendations for action in the event of natural disasters such as floods or droughts.

In the transformation of cities into smart cities, precipitation data also plays an important role [1–3]. Heavy rainfall in urban areas can lead to sewer system failures, local flooding and flash floods [4–6]. These events pose a significant risk to people and infrastructure. Therefore, it is important to take preventive measures to reduce the impact of heavy rainfall in these areas. In urban environments, there are special circumstances that make precipitation measurement particularly challenging. The unique characteristics of urban areas, such as sealed surfaces, changing land use patterns and complex topography, create microclimates that influence precipitation distribution. Conventional meteorological stations designed for open landscapes often cannot adequately capture the variations in precipitation intensity and patterns

in urban areas. Accurate and locally resolved precipitation data is needed to warn residents and emergency services of extreme weather situations at an early stage. This requires dense sensor network covering the city area [7,8]. In addition to high-precision measuring devices, this also requires low-cost sensors [9] that can be deployed in a dense IoT network. Low-cost sensors have the potential to enable the development of dense sensor networks and provide precipitation data that can improve our understanding of local weather patterns and support effective urban planning and management. For this purpose, a new type of inductive rain sensor was introduced in [10]. The sensor is intended to allow rain data to be captured in real time and transmitted wirelessly over LoRaWAN. Furthermore, the sensor is energy self-sufficient, resulting in low maintenance requirements. As no collecting tray is used, the sensor measures precipitation from the first drop, so there is no time delay, which can be particularly important during heavy rainfall events. The sensor principle was evaluated in laboratory conditions and proved to be suitable for the application. Field tests are now required for further development of the sensor. This data can then be used to create a signal processing algorithm. The algorithm should be developed in two stages. First, peak detection is required to identify the incoming drops.

* Corresponding author.

E-mail addresses: christoph.clemens@hs-ruhrwest.de (C. Clemens), annette.jobst@hs-ruhrwest.de (A.E. Jobst), mario.radschun@hs-ruhrwest.de (M. Radschun), joerg.himmel@hs-ruhrwest.de (J. Himmel), olfa.kanoun@etit.tu-chemnitz.de (O. Kanoun).

<https://doi.org/10.1016/j.measurement.2024.114286>

Received 8 November 2023; Received in revised form 22 January 2024; Accepted 5 February 2024

Available online 8 February 2024

0263-2241/© 2024 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

In the second step, the precipitation from the drops must be determined. The algorithm should be designed in such a way that it can execute on a microcontroller. Accordingly, it should be as simple an approach as possible that requires little computing power and energy. To achieve these goals, an algorithm for drop detection and two approaches for precipitation calculation are proposed in this paper.

The work presented here is embedded in a current research project whose aspired result is the prediction of heavy rainfall and flash floods in urban areas [10]. The AI used for this purpose requires real time measurement data to locally predict heavy rain events and to mark the area at risk of flooding [10].

1.1. State of the art

There are already many approaches for the detection of impulses [11,12]. Mostly, these are used to suppress impulse noise. Median filters are used for the detection. The deviation of the signal is considered and then a threshold value is used to determine whether it is an impulse or not. However, since the signal triggered by a drop is not a single peak but a damped harmonic oscillation, these algorithms are not applicable.

In [13] a model is presented which detects damped harmonic oscillations in highly noisy signals. Variational mode decomposition is used to decompose a complex signal into intrinsic functions [14]. After that the kurtosis index is used [15]. The index is sensitive to impact components of the signal and describes the ratio between the fourth order moment and the square of the second order moment. To determine if it is a damped oscillation or just a single pulse, a cross correlation is performed. This method is suitable for detecting damped harmonic oscillations in noisy signals and would be a good approach for drop detection. However, this method is time consuming and requires high computational power to perform the steps required for it.

A piezoelectric low-cost rain sensor that operates on the principle of detecting the impact of a drop is presented in [16]. When a drop hits the sensor, a damped harmonic oscillation is generated due to the piezoelectric effect. The signal is recorded via a sound card which is embedded in a computer. To detect the drops, the local maximum is taken into account. When it reaches a certain threshold, it is marked as a drop. After the drops are detected, the information is fed into an AI that has been trained using a reference measurement system. The AI then estimates the amount of precipitation. The described approach would be very applicable to the inductive rain sensor signals, but has the following limitations. The detection of raindrops does not distinguish whether it is really a raindrop or a disturbance that may be caused by environmental factors such as wind. In addition, due to the use of an AI, this approach is very computationally intensive and therefore cannot be used. The use of an AI is not based on a physical model, nor does it allow for causality testing, so errors can occur. Especially in the case of extraordinary rain events, which could not be trained sufficiently.

2. Materials and methods

2.1. Inductive rain sensor

The inductive sensor for which the signal processing is developed has

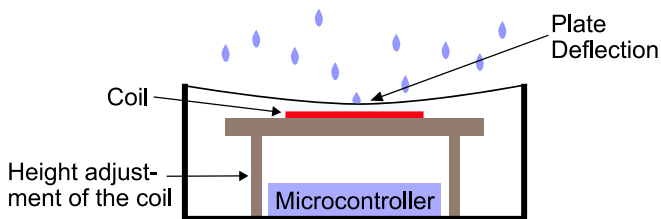


Fig. 1. Schematic view of the inductive rain sensor with drops hitting the plate and causing a deflection.

already been described in [10]. The design is shown schematically in Fig. 1. The sensor consists of an electrically conductive plate, a coil which is operated in a resonant circuit and a microcontroller. The coil is located under the plate so that an eddy current is induced in the plate. The distance between the plate and the coil can be varied so that the highest possible sensitivity is achieved without the plate touching the coil due to deflection. When raindrops fall on the plate, it is deflected. This influences the inductance of the coil and leads to a change in the oscillating circuit frequency. This frequency is measured with the microcontroller using the pulse train.

The plate, on which the rain hits, is clamped on all sides and represents a vibration-damped system that can be described by a mass-spring-damper arrangement. The measurement effect occurs when a raindrop hits the plate. The drop acts as an external force on the plate and generates a free damped oscillation. Assuming that the duration of the force applied by the drop is very short in relation to the time constant of the mass-spring-damper system, this can be regarded as a force pulse. The general solution for a 2nd order system for an impulse excitation is as follows.

$$x(t) = A_0 \cdot \sin(\omega t + \varphi) \cdot e^{-\delta t} \quad (1)$$

Where the constants A_0 and φ describe the initial amplitude of the damped oscillation and the phase shift. The decay time of the oscillation depends only on the damping factor δ of the system. Fig. 2 (a) shows this behavior for different damping factors. The damping of the system can be determined via the change of the amplitude over time.

$$-\delta \cdot t_1 = \ln\left(\frac{A_1}{A_0}\right) \quad (2)$$

However, practical tests are necessary to determine the damping. For this purpose, experiments were carried out in the laboratory. Fig. 2 (b) shows the bandpass filtered signal triggered by a drop. To determine the damping factor, 10 drops were analyzed and the average damping factor was calculated.

2.2. Raindrop detection

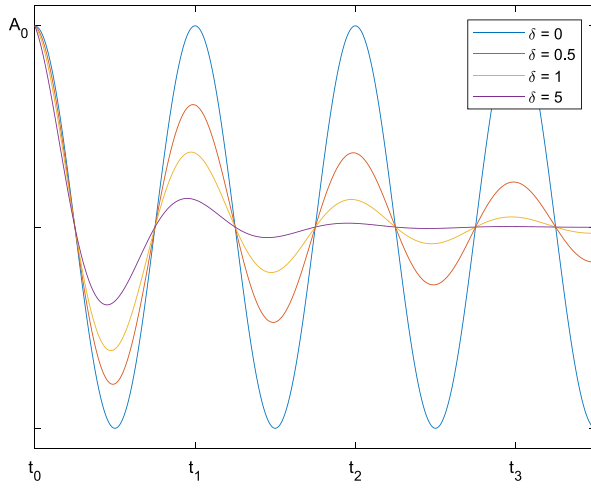
The impulse response of the system can now be used to develop an algorithm that can detect incoming raindrops. First, the signal is pre-processed. For this purpose, a bandpass filter is designed to eliminate the offset and suppress disturbances that are not in the range of the plate's eigenfrequency. Since it is expected that the eigenfrequency may change due to drops on the plate changing the mass of the system during a rain event, the filter was designed with a bandwidth of $\pm 20\%$ of the eigenfrequency. Filtering the offset of the signal also has the advantage that the base frequency of the oscillating circuit, which is dependent on the temperature and the associated geometric change of the housing and coil, no longer has any influence.

In order to be able to identify drops, peak values in the frequency signal are first determined. For this purpose, local maxima are marked in the signal, which have a minimum value of 1000 Hz. This corresponds to five times the resolution of the frequency measurement. The marked maxima have a minimum time interval of 15 ms. Taking into account the damping and the amplitude, the amplitude function of a damped harmonic oscillation is now applied to the signal. The marked maximum is the amplitude A_0 . The time $t_0 = 0$ is the starting point of the decaying harmonic.

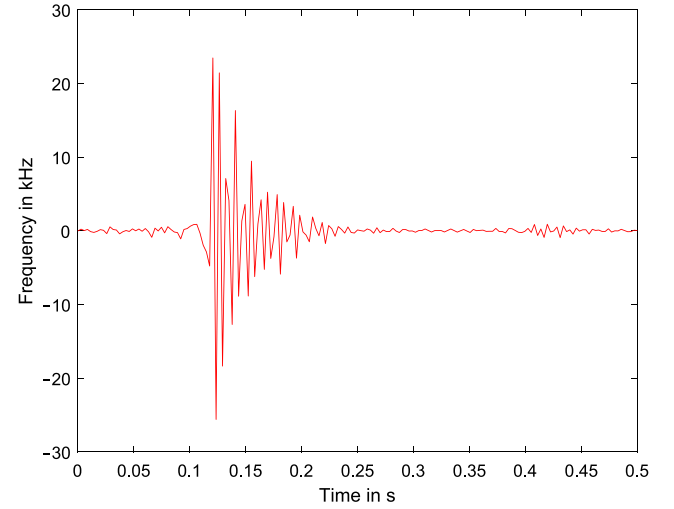
$$x(t) = A_0 \cdot e^{-\delta t} \quad (3)$$

$$x(t_0) = A_0 \cdot e^{-\delta t_0} = A_0 \quad (4)$$

With the help of the two following maxima at time t_1 and t_2 , it is now checked whether it is an oscillation or a single peak. For this purpose, the difference ΔA_+ between these maxima and the amplitude function (3) is calculated.



(a)



(b)

Fig. 2. (a) Impulse response of a 2nd order system for different damping factors. (b) Signal triggered by a drop.

$$\Delta A_+ = |(A_0 \bullet e^{-\delta t_1} - A_1)| + |(A_0 \bullet e^{-\delta t_2} - A_2)| \quad (5)$$

If the difference is less than 15 %, it is classified as a damped oscillation and the first maximum value is marked as a drop.

$$\Delta A_+ < 0.15 \bullet (A_0 \bullet e^{-\delta t_1} + A_0 \bullet e^{-\delta t_2}) \quad (6)$$

Fig. 3 shows a damped oscillation and the applied ideal amplitude function with the maxima A_0 , A_1 and A_2 marked.

To enhance the detection rate, the algorithm was expanded to include negative parts of the oscillation in addition to the positive parts. For this purpose, the negative oscillation peaks and the deviation from the amplitude function are determined using the same method as for the positive peaks.

Besides the raindrops, the sensor is also exposed to other factors that can cause the plate to oscillate. The major influence, similar to rain sensors that use a collection principle, is the wind [17]. The wind excites the plate with the vortex effect and also triggers an oscillation. The eigenfrequency excited by wind is the same as that of a drop hitting the plate, so this disturbance cannot be removed by a filter. The characteristics of excitation by wind and excitation by a drop are different. In

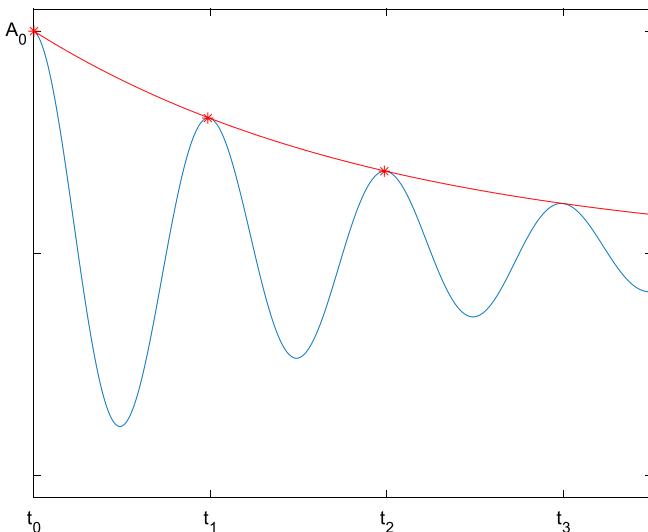


Fig. 3. Damped harmonic oscillation with local maxima.

the case of wind, an increasing harmonic oscillation is initially observed and, after reaching a maximum amplitude, a decaying harmonic oscillation succeeds. In the case of impulsive excitation by a drop, the period of the rising oscillation is omitted and only a decaying harmonic is observed. To distinguish this and prevent false drop detection, a second criterion is added to the algorithm in the form of a rising harmonic.

$$x(t) = A_0 \bullet e^{\delta t} \quad (7)$$

Now the two amplitudes preceding the marked amplitude A_0 are used to form a difference to the rising harmonic.

$$\Delta A_- = |(A_0 \bullet e^{\delta(t_1-t_0)} - A_{-1})| + |(A_0 \bullet e^{\delta(t_2-t_0)} - A_{-2})| \quad (8)$$

If the difference deviates by more than 85 %, the maximum A_0 remains marked as a drop. If the deviation is less, it is a false detection and the marking is removed. Accordingly, the drop is only detected if both criteria are met.

2.3. Precipitation calculation

After determining the number of drops, the resulting amount of precipitation is to be calculated. Two different approaches are being pursued for this purpose.

The first approach is based on a statistical distribution of the drop size during a rain event. The size distribution of raindrops has already been described in other publications and is given as a gamma function [18–21].

$$N(D) = N_0 D^\mu \bullet e^{-\Lambda \bullet D} \quad (9)$$

Where μ is the shape parameter, D is the possible drop size from 0 to 8 mm, N_0 is the value for $N(D)$ when $D = 0$ and Λ is the slope parameter, which depends on the median volume diameter D_0 . This parameter varies with the amount of rainfall R .

$$\Lambda = \frac{3.67 + \mu}{D_0} \quad (10)$$

$$D_0 = \epsilon \bullet R^\delta \quad (11)$$

Since the distribution of drop size varies with precipitation intensity and the distribution shifts to larger drops during heavy rain events, four different curves were defined and are shown in Fig. 4. The gamma function is now used to divide the drop sizes into categories and calculate their percentage share. The categories cover the range 0–8 mm

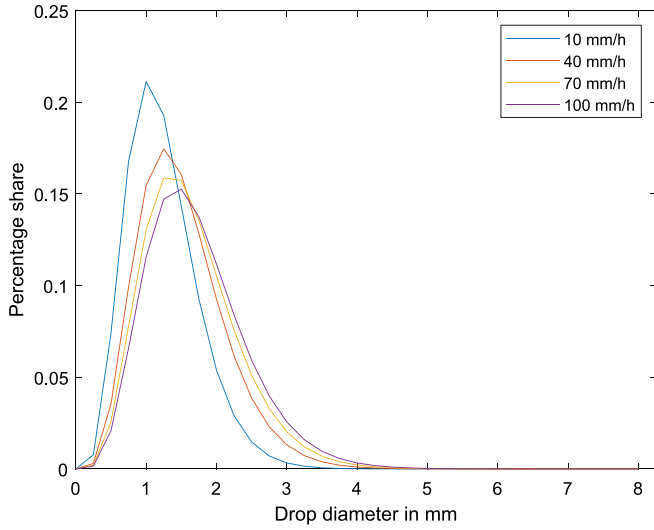


Fig. 4. Gamma function of drop size distribution for different precipitation intensities.

in 0.5 mm steps. The algorithm then decides which of the four curves is used for the calculation based on the number of drops detected per minute. This assumes that the more drops are detected, the greater the rain intensity. Since this is not always the case, inaccuracies in precipitation calculations can occur with this method. After categorization, precipitation is calculated in $\frac{mm}{min}$ and $\frac{mm}{h}$.

The second approach considers the frequency peak triggered by the raindrop. The detected drops are categorized according to their peak height and each category is assigned a specific drop size. The amount of precipitation is then calculated. It is important to keep in mind that the position where the drop hits the plate affects the height of the peak. A smaller drop hitting the plate in the center can trigger the same deflection and therefore the same peak height as a larger drop hitting the plate near the edge. A finite element simulation to determine this effect has already been described in [10]. The simulation is now being adapted and improved to better determine the influence of the impact position. Fig. 5 (a) shows the deflection caused by different drop sizes. The boxplot shows the deflection range in dependence of the position where the

drop hits the plate. An equal distribution can be assumed for the impact points of the drops on the plate. The median value of the deflection caused by a drop is used for categorization. For the calculation it is necessary to convert the change in the distance between coil and plate into a change in frequency. The eigenfrequency of the electromagnetic resonant circuit depends on the inductance (L) and the capacitance (C) and can be represented as follows.

$$\omega = \frac{1}{\sqrt{L \cdot C}} \quad (12)$$

The eddy current induced in the plate by the magnetic field of the coil causes a mutual inductance (L_M) which varies with the distance between the coil and the plate. The mutual inductance reduces the inductance of the coil (L_{ind}), which changes the frequency of the resonant circuit.

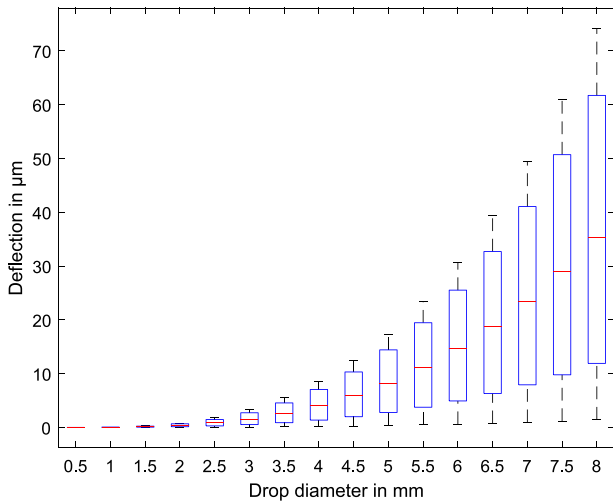
$$\omega = \frac{1}{\sqrt{(L_{ind} - L_M(d)) \cdot C}} \quad (13)$$

The frequency decreases exponentially with the decreasing mutual inductance but can be considered linear for small changes. The same applies to the eddy currents and the mutual inductance, which decrease exponentially with distance between plate and coil, but can be considered linear for the close range and small changes in distance [10]. Therefore, the change in frequency as a function of the change in distance can be assumed to be linear for small changes. The distance change Δd is given in μm and the frequency change $\Delta \omega$ in Hz.

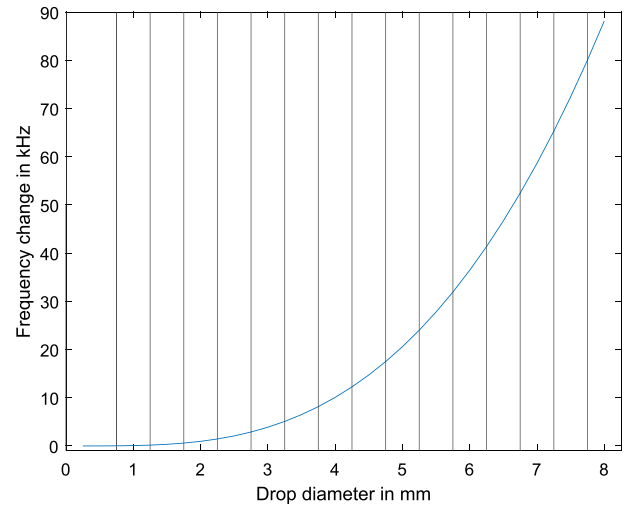
$$\Delta \omega = 2500 \frac{Hz}{\mu m} \cdot \Delta d \quad (14)$$

The median of the deflection was converted into a frequency change. The drop sizes were then categorized and corresponding frequency changes for the categories were determined. The categories are plotted in Fig. 5 (b) and the frequency values are shown in Table 1.

According to this principle, each detected drop is assigned to a category depending on its peak height. Then the values are added over one minute to determine the amount of precipitation in $\frac{mm}{min}$ respectively in $\frac{mm}{h}$. This approach may result in drops being assigned to the wrong category, which can lead to inaccuracies. It is to be expected that the drops will be assigned to a lower category rather than a higher one. The reason for this is that larger drops have a larger dispersion due to the position of their impact. This can be seen in the boxplot. Therefore, this



(a)



(b)

Fig. 5. (a) Boxplot of the deflection caused by different drop sizes at different positions. (b) Frequency changes triggered by deflection and resulting categorization of drop sizes in 0.5 mm steps.

Table 1

Categorization of drop size with corresponding range of frequency change.

Drop size in mm	Frequency change in kHz
0.5	<0.75
1.0	0.75–1.25
1.5	1.25–1.75
2.0	1.75–2.25
2.5	2.25–2.75
3.0	2.75–5.0
3.5	5.0–8.5
4.0	8.5–12.5
4.5	12.5–17.5
5.0	17.5–24.0
5.5	24.0–32.0
6.0	32.0–41.5
6.5	41.5–52.5
7.0	52.5–65.5
7.5	65.5–80.0
8.0	>80.0

method probably leads to an underestimation of the precipitation.

The final calculation of the precipitation in $\frac{mm}{h}$ is the same for both approaches. It should be noted that in reality falling raindrops are usually not spherical [22]. However, the previously determined diameters refer to an equivalent volume of a sphere [18]. Therefore, the formulas of a sphere can be used to determine the volume of a drop [23]. The density ρ of water can be used to calculate the mass of each drop. In order to determine the precipitation with a temporal resolution of one minute, the mass of the drops is summed up over one minute. The summation is shown in equation (15), where x is the number of drops per minute and r is the radius of the drops.

$$m_{min} = \sum_{i=1}^x \frac{4}{3} \cdot \pi \cdot r_i^3 \cdot \rho \quad (15)$$

The area of the sensor on which the rain hits has a diameter of 20 cm, so the value must be multiplied by a factor to obtain the amount of precipitation per square meter.

$$m_{ges} = m_{min} \cdot \frac{1}{\pi \cdot 0.1^2} \quad (16)$$

Since one kilogram corresponds to 1 mm of precipitation per square meter, this value is equivalent to an indication in $\frac{mm}{min}$. Multiplying m_{ges} by 60 gives the precipitation in $\frac{mm}{h}$.

2.4. Measurement setup

The function of the sensor was first demonstrated in the laboratory [10]. However, since it is very complex to simulate rain in the laboratory and it is impossible to take all additional influences into account, the sensor was used in the field to record frequency data during real rain events. Precipitation data were recorded in parallel with a reference measuring device. These measurement data form the basis of the signal processing.

A weighing pluviometer of the rain[e] series from Lambrecht meteo [24], which is also used as a standard measuring system by the German Weather Service (DWD), was used as a reference measuring system. A pluviometer is a weighing precipitation sensor. The precipitation is collected by means of a cylindrical container and then formed into predefined drops by a funnel and recorded by a weighing scale. The rain [e] is a self-emptying system. A tipping mechanism ensures that the chamber of the weighing scale empties itself, which enables continuous measurement. The specifications of the system are given in Table 2. The specifications refer to the amount of precipitation on one m^2 , where the collecting area of the sensor is only 200 cm^2 .

In a report of the DWD from 2020 [25], the rain[e] was compared with a manual precipitation gauge according to Hellmann. The

Table 2

Specifications of the rain[e] sensor.

Intensity range	Intensity resolution	Precision
0–20 $\frac{mm}{min}$	0,001 $\frac{mm}{min}$	$\pm 0.1 mm, \pm 1\% < 6 \frac{mm}{min}, \pm 2\% > 6 \frac{mm}{min}$

Hellmann precipitation gauge works according to a simple accumulation principle. The precipitation is collected in a container and after 24 h the amount is read [26]. Thus, the precipitation amount of one day can be determined. The values of the rain[e] summed up over one day were within the tolerance range of 7 % deviation from the absolute value of the Hellmann sensor specified by the World Meteorological Organization (WMO). This makes the rain[e] a suitable reference instrument for the acquisition of “live” precipitation data.

Since the measurement is a reference measurement, the absolute value of the precipitation plays only a minor role. It was important that the reference measuring instrument and the inductive rain sensor are subject to the same measuring conditions. Nevertheless, the recommendations of the WMO for the measurement of precipitation were followed for the measurement site [27]. Accordingly, the measuring devices should be set up in an open area so that the precipitation is not affected by buildings. The WMO recommends that the distance to obstacles should be twice the height of the obstacle. The measuring devices should also be placed at a height of 1 m. Fig. 6 (a) shows the location of the sensors. They are placed on a flat surface at a distance of one meter from each other. The sensors were each mounted on a metal pole and fixed with a stand. The distances to surrounding obstacles are shown schematically in Fig. 6 (b). The wind direction is mainly from the west. No additional precautions have been taken to reduce wind or other potential disturbances.

The rain[e] data is read out every minute. The measured value is in $\frac{mm}{h}$. The frequency of the oscillating circuit of the inductive sensor is continuously recorded. In order to obtain comparable data, the continuously recorded data is divided into intervals of one minute so that it can also be evaluated every minute.

3. Results and discussion

3.1. Drop detection

The raindrop detection algorithm has now been applied to real measurement data. Fig. 7 (a) shows how the algorithm works. The local maximum was marked and an increasing and decreasing harmonic was applied over the signal. In this case, the local maximum is followed by a damped harmonic oscillation. The deviation from the applied falling harmonic is accordingly small and thus the first criterion is fulfilled. Furthermore, it can be seen that there is no rising harmonic before the local maximum. Therefore, the deviation from the rising harmonic is large, which means that the second criterion is also satisfied. Thus, both criteria of the algorithm are fulfilled and the maximum is marked as a drop. (b) shows a marked maximum that meets the first criterion due to the falling harmonic after the peak, but was identified as a false detection by the second criterion due to the rising harmonic.

To verify the algorithm, the drops occurring in the signal from 10 different rain events were counted and marked manually. For this purpose, the drops were counted for one minute and then the algorithm was applied. The results are shown in Table 3. The algorithms always achieved a detection rate of over 85 %. The average rate over 10 events is 89.25 %. Factors that can lead to a drop not being detected are a possible superposition of an excitation by a drop and by wind and the time interval at which the drops hit the plate. As there is a minimum time of 15 ms between two detected drops, it is possible that drops are not detected if a second drop hits the plate less than 15 ms after the first drop. It was also determined in a simulation that the smallest possible detectable drop size is 0.2 mm. At this drop size, the average deflection of the plate



Fig. 6. (a) Inductive rain sensor and reference sensor rain[e]; (b) Schematic of the measurement site.

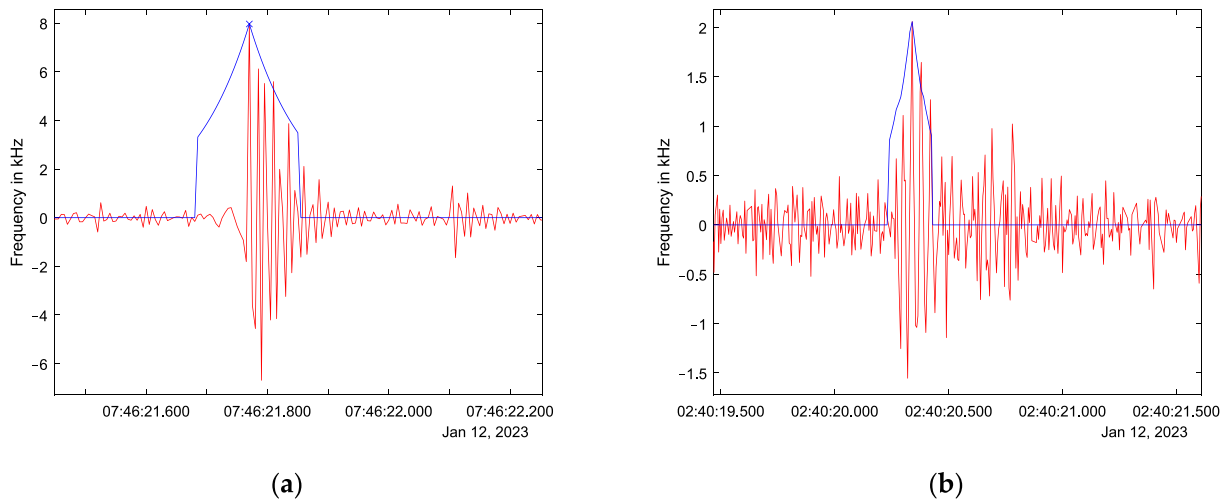


Fig. 7. (a) A drop correctly marked by the algorithm; (b) Disturbances originally marked as drops and identified as fault detections by the second criterion.

Table 3
Accuracy of the drop detection algorithm.

Event	Manually marked drops	Drops detected by the algorithm	Detection rate
1	188	165	87.77 %
2	60	51	85.00 %
3	79	75	94.94 %
4	128	115	89.84 %
5	185	169	91.36 %
6	116	99	85.34 %
7	219	195	89.04 %
8	53	46	86.79 %
9	97	90	92.78 %
10	106	95	89.62 %

is less than $0.4 \mu\text{m}$ and therefore causes a frequency change below 1000 Hz, which is no longer recognized as a drop by the algorithm.

3.2. Precipitation calculation

Now the two approaches described are applied to calculate the precipitation from the detected drops.

Fig. 8 shows the precipitation during a heavy rain event that occurred on June 22, 2023 in Western Germany [28]. The reference data measured by the rain[e] and the calculated precipitation amounts of the

two approaches are shown. Each measured value represents one minute. The total amount of rain for this event is shown in Table 4 under Event 1. Even so the deviation of the total rain amount for the gamma function is only 11.06 %, it can be seen that the values for each minute deviate significantly from the reference data. This is due to the fact that even at intensities of over $40 \frac{\text{mm}}{\text{h}}$, only approx. 200 drops per minute hit the sensor. Since these are assigned to the respective category on a percentage basis, errors occur in the calculation in the minute range with so few drops. When calculating the amount of rain for the entire rain event, the number of drops is significantly higher and therefore an evaluation based on statistics performs better. This also applies to the precipitation events listed in the table below. In the table, only the precipitation quantity of an entire rain event was compared. The reason for this is that the rain[e] has a certain delay time between the raindrop hitting the collection funnel and being recorded by the weighing principle. This varies depending on the amount of rain. During light rain, it takes longer for the water to reach the outlet of the funnel and then fall onto the scale, as the drops do not immediately run off the inside of the collection funnel. With the measuring principle of the low-cost sensor, the drop is detected as soon as it falls onto the plate. This makes a minute-by-minute comparison of the two sensors difficult. The average deviation for the 20 events listed is 12.72 %. Due to the low precipitation amount of only 2.95 mm at Event 2, a high percentage deviation results, although the absolute deviation is not so large. Neglecting the outlier of event 2, the

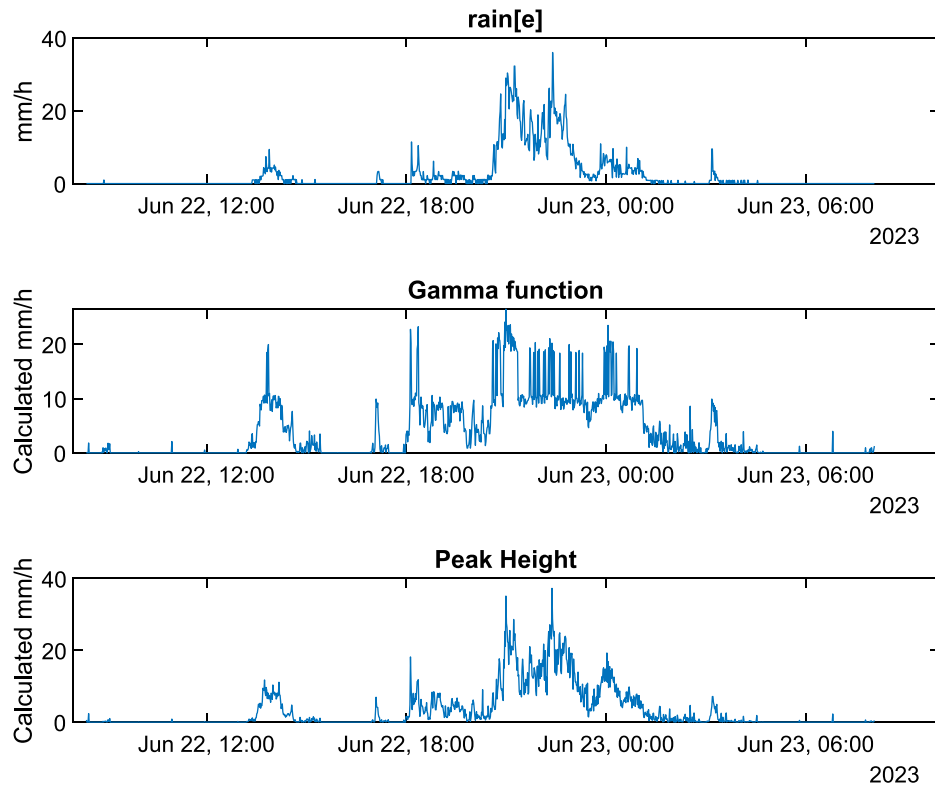


Fig. 8. Reference data measured by the rain[e]. Precipitation calculated with the gamma function and the peak height method.

Table 4

Deviation of the two approaches compared to the reference measurement for 10 rain events.

Event	Total rain in mm rain[e]	Total rain in mm Gamma function	Deviation	Total rain in mm Peak height	Deviation
1	56.31	62.54	11.06 %	52.59	−6.61 %
2	2.95	4.22	43.39 %	1.95	−33.65 %
3	9.49	7.30	−23.05 %	9.52	−6.16 %
4	3.56	4.13	16.17 %	2.99	−15.91 %
5	14.96	16.09	7.56 %	12.73	−14.87 %
6	20.80	22.39	7.64 %	16.86	−18.92 %
7	2.62	3.19	21.94 %	1.95	−25.57 %
8	10.53	9.96	−5.41 %	9.07	−13.83 %
9	7.86	8.63	9.78 %	6.84	−12.96 %
10	18.32	20.64	12.64 %	16.77	−8.45 %
11	6.52	5.65	−13.51 %	6.23	−4.48 %
12	7.72	6.66	−13.78 %	6.97	−9.70 %
13	10.94	10.81	−1.15 %	9.54	−12.81 %
14	10.48	9.69	−7.48 %	9.13	−12.87 %
15	26.92	29.37	9.10 %	25.77	−4.29 %
16	27.97	24.51	−13.36 %	24.79	−11.35 %
17	5.60	5.25	−6.17 %	4.92	−12.18 %
18	7.67	8.69	13.83 %	7.25	−5.48 %
19	13.98	14.60	4.41 %	11.73	−16.11 %
20	6.04	6.82	12.97 %	5.66	−6.29 %

deviation is only 11.11 %.

The approach using peak height shows better correlation in calculating the amount of rain per minute. The total deviation for the first event is only −6.61 %. Considering the events listed in the table. The average deviation is 12.62 %. Excluding Event 2, it is only 11.51 %.

There are several possible sources of error that explain the deviations from the reference system. Possible sources of error for the first approach are the parameters of the gamma function. This function estimates the drop size distribution based on parameters that differ for

different types and amounts of precipitation, which can lead to inaccuracies. In addition, only four different rain intensities were calculated based on the number of detected drops. This can lead to errors as these curves cannot represent all rain intensities. Also, as already mentioned in chapter 2.3, a large number of drops does not necessarily mean that the rain intensity is high.

For the second approach it should be noted that all values are below the reference values, which confirms the assumption that the drops are more often classified in a smaller category. Other sources of error are wind. Deviations can occur due to both the wind force and the wind direction. The wind force influences the speed of the droplets and therefore the peak height, which can lead to incorrect categorization. Since the sensor has a slight inclination so that the water can run off, not only the wind velocity but also the wind direction is relevant. If the wind comes from the front, more drops hit the sensor. However, if it comes from behind, there are fewer. This influences both approaches as the number of detected drops depends on the wind. In addition, the drop detection influences the calculated amount of rain, as not all drops are detected.

4. Conclusions

The drop detection algorithm shows a high detection rate of 89.25 %. In addition, external disturbances such as those caused by wind are largely suppressed. Furthermore, the algorithm is easy to apply, making it suitable for use in an energy-autonomous system without much computing power.

The approaches for calculating the amount of precipitation offer advantages and disadvantages. For example, the statistical approach using the drop size distribution with the help of a gamma function has only a deviation of 11.11 % to the reference measurement. However, in the real-time application, strong deviations show up in an evaluation accurate to the minute. This behavior could be improved by depositing a larger number of gamma functions so that the drop size distribution can be adjusted more dynamically.

The approach using peak height has slightly worse correlation in total rainfall, but shows better performance in the real-time application. It should be noted that this approach always resulted in an underestimation of rainfall. This can possibly be corrected if the frequency ranges of the categories are adjusted. In addition, it would be conceivable to expand the categories to include larger drops, since drops can overlap and thus two simultaneously impacting drops are classified as one large drop that exceeds the physically possible size of a drop. Another possibility for improvement would be an additional wind measurement. This would allow the influence of wind strength and wind direction on the sensor to be measured and the peak heights to be adjusted accordingly [29].

The majority of rain drops have a size between 1 mm and 2 mm. To achieve a better result, the size of the categories in this range could be refined from 0.5 mm to <0.25 mm. The deviation resulting from the evaluation of the drop detection can also be used to correct the calculated values. Since the sensor is to be used in a dense sensor network, which also includes high-precision sensors, there is also the possibility of correcting data and thus achieving an improvement. Furthermore, it remains to be noted that the accuracy for small amounts of rain is not so high. However, since the system is mainly intended to detect heavy rainfall, this is negligible.

From the results, it is shown that both approaches are suitable for the application. They require little computing power and can execute on a microcontroller. Therefore, both approaches should be pursued and improved with the measures described. In future work, the algorithms should now be transferred to the microcontroller so that precipitation data can be calculated in real time.

CRedit authorship contribution statement

Christoph Clemens: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Annette Elisabeth Jobst:** Validation, Writing – review & editing. **Mario Radschun:** Validation, Writing – review & editing. **Jörg Himmel:** Conceptualization, Funding acquisition, Project administration, Supervision, Validation. **Olfa Kanoun:** Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Christoph Clemens reports financial support was provided by Federal Ministry of Education and Research (BMBF). Grant ID: 13N15556–13N15560. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] S. Banara, T. Singh, A. Chauhan, IoT based weather monitoring system for smart cities: a comprehensive review, in: International Conference for Advancement in Technology (ICONAT), Goa, India, 2022, pp. 1–6, <https://doi.org/10.1109/ICONAT53423.2022.9726106>.
- [2] A.-U. Rahman, S. Abbas, M. Gollapalli, R. Ahmed, S. Aftab, M. Ahmad, M.A. Khan, A. Mosavi, Rainfall prediction system using machine learning fusion for smart cities, *Sensors* 22 (2022) 3504, <https://doi.org/10.3390/s22093504>.
- [3] Y. Zhou, A. Sharma, M. Masud, G.S. Gaba, G. Dhiman, K.Z. Ghafoor, M.A. AlZain, Urban rain flood ecosystem design planning and feasibility study for the enrichment of smart cities, *Sustainability* 13 (2021) 5205, <https://doi.org/10.3390/su13095205>.
- [4] L. Cea, P. Costabile, Flood risk in urban areas: modelling, management and adaptation to climate change. A review, *Hydrology* 9 (2022) 50, <https://doi.org/10.3390/hydrology9030050>.
- [5] L.G. Lanza, A. Cauteruccio, E. Chinchella, Opportunistic rain sensors and flood modelling to assess the risk of failure of surface drainage in urban areas, in: EGU General Assembly 2023, Vienna, Austria, 24–28 Apr 2023, EGU23-9567, 2023, <https://doi.org/10.5194/egusphere-egu23-9567>.
- [6] J.A. Marengo, E. Alcantara, A.P. Cunha, M. Seluchi, C.A. Nobre, G. Dolif, D. Goncalves, M. Assis Dias, L.A. Cuartas, F.F. Bender, Flash floods and landslides in the City of Recife, Northeast Brazil after heavy rain on May 25–28, 2022: causes, impacts, and disaster preparedness, *Weather Clim. Extrem.* 2023 (39) (2022) 100545, <https://doi.org/10.1016/j.wace.2022.100545>.
- [7] R. Krüger, P. Karrasch, A. Eltner, Calibrating low-cost rain gauge sensors for their applications in IoT infrastructures to densify environmental monitoring networks, *Geosci. Instrum. Method. Data Syst. Discuss.* [preprint], doi: 10.5194/gi-2023-7.
- [8] D.K. Singh, R. Sobti, A. Jain, P.K. Malik, D. Le, LoRa based intelligent soil and weather condition monitoring with internet of things for precision agriculture in smart cities, *IET Commun.* 16 (2022) 604–618, <https://doi.org/10.1049/cmu2.12352>.
- [9] B. Verbeiren, J. Lemmens, Exploring the added value of low-cost sensors via citizen observatories for peak rainfall monitoring in cities (Case study: Brussels), EGU General Assembly 2023, Vienna, Austria, 24–28 Apr 2023, EGU23-16721, doi: 10.5194/egusphere-egu23-16721.
- [10] C. Clemens, A. Jobst, M. Radschun, J. Himmel, O. Kanoun, M. Quirnbach, Development of an inductive rain gauge, *Sensors* 22 (2022) 5486, <https://doi.org/10.3390/s22155486>.
- [11] T. Chen, H.R. Wu, Adaptive impulse detection using center-weighted median filters, *IEEE Signal Process. Lett.* 8 (1) (2001) 1–3, <https://doi.org/10.1109/97.889633>.
- [12] S. Zhang, M.A. Karim, A new impulse detector for switching median filters, *IEEE Signal Process. Lett.* 9 (11) (2002) 360–363, <https://doi.org/10.1109/LSP.2002.805310>.
- [13] S. Wang, B. Lu, Detecting the weak damped oscillation signal in the agricultural machinery working environment by vibrational resonance in the duffing system, *J. Mech. Sci. Technol.* 36 (2022) 5925–5937, <https://doi.org/10.1007/s12206-022-1109-3>.
- [14] K. Dragomiretskiy, D. Zosso, Variational mode decomposition, *IEEE Trans. Signal Process.* 62 (3) (2014) 531–544, <https://doi.org/10.1109/TSP.2013.2288675>.
- [15] J. Zhong, D. Wang, J. Guo, D. Cabrera, C. Li, Theoretical investigations on kurtosis and entropy and their improvements for system health monitoring, *IEEE Trans. Instrum. Meas.* 70 (2021) 1–10, <https://doi.org/10.1109/TIM.2020.3031125>.
- [16] A. Antonini, S. Melani, A. Mazza, L. Baldini, E. Adiroso, A. Ortolani, Development and calibration of a low-cost, piezoelectric rainfall sensor through machine learning, *Sensors* 22 (2022) 6638, <https://doi.org/10.3390/s22176638>.
- [17] J. Kochendorfer, R. Rasmussen, M. Wolff, B. Baker, M.E. Hall, T. Meyers, S. Landolt, A. Jachcik, K. Isaksen, R. Brækkan, R. Leeper, The quantification and correction of wind-induced precipitation measurement errors, *Hydrol. Earth Syst. Sci.* 21 (2017) 1973–1989, <https://doi.org/10.5194/hess-21-1973-2017>.
- [18] C.W. Ulbrich, Natural variations in the analytical form of the raindrop size distribution, *J. Clim. Appl. Meteorol.* 22 (1983) 1764–1775, <http://www.jstor.org/stable/26181251>.
- [19] D. Atlas, C. Ulbrich, Drop size spectra and integral remote sensing parameters in the transition from convective to stratiform rain, *Geophys. Res. Lett.* (2006) 33, <https://doi.org/10.1029/2006GL026824>.
- [20] E.F. Campos, I. Zawadzki, M. Petitdidier, W. Fernandez, Measurement of raindrop size distributions in tropical rain at Costa Rica, *J. Hydrol.* 328 (2006) 98–109, <https://doi.org/10.1016/j.jhydrol.2005.11.047>.
- [21] K.V. Beard, V.N. Bringi, M. Thurai, A new understanding of raindrop shape, *Atmos. Res.* 97 (2010) 396–415, <https://doi.org/10.1016/j.atmosres.2010.02.001>.

- [22] B.K. Seela, J. Janapati, P.-L. Lin, P.K. Wang, M.-T. Lee, Raindrop size distribution characteristics of summer and winter season rainfall over north Taiwan, *J. Geophys. Res. Atmos.* (2018) 123, <https://doi.org/10.1029/2018JD028307>.
- [23] M. Angulo-Martinez, A.P. Barros, Measurement uncertainty in rainfall kinetic energy and intensity relationships for soil erosion studies: an evaluation using PARSIVEL disdrometers in the Southern Appalachian Mountains, *Geomorphology* 228 (2015) 28–40, <https://doi.org/10.1016/j.geomorph.2014.07.036>.
- [24] Lambrecht meteo Homepage, Available online: <https://www.lambrecht.net/produkte/niederschlag/wiegender-niederschlagssensor-rain-e> (accessed on 03 March 2022).
- [25] L. Hannak, S. Brinckmann, Parallelmessungen an deutschen Klimareferenzstationen, in: 253rd ed.; Self-published by the German Weather Service, Offenbach, Germany, 2020; pp. 42–54 ISBN 978-3-88148-523-4 Online Available: https://www.dwd.de/DE/leistungen/pbfb_verlag_berichte/pdf_einzelbaende/253_pdf.pdf?_blob=publicationFile&v=4 (accessed on 18 September 2023).
- [26] A. Kotowski, A. Dancewicz, B. Kaźmierczak, Accuracy of measurements of precipitation amount using standard and tipping bucket pluviographs in comparison to hellmann rain gauges, *Environ. Prot. Eng.* 37 (2) (2011, 2011,) 23–34.
- [27] World Meteorological Organization (WMO), Guide to Instruments and Methods of Observation, 2021 edition, WMO, Geneva, Swiss, 2021, p. 226 ISBN 978-92-63-10008-5.
- [28] Tagesschau Homepage, Schwere Unwetter ziehen über Deutschland, Available online: <https://www.tagesschau.de/wetter/unwetter-deutschland-144.html> (accessed on 19 September 2023).
- [29] A. Cauteruccio, M. Stagnaro, L.G. Lanza, P.-W. Chan, Adjustment of 1 min rain gauge time series using co-located drop size distribution and wind speed measurements, *Atmos. Meas. Tech.* 16 (2023) 4155–4163, <https://doi.org/10.5194/amt-16-4155-2023>.